

Machine-Learning Interatomic Potentials for Long-Range Systems

Yajie Ji,^{1,*} Jiuyang Liang,^{1,2,†} and Zhenli Xu^{1,3,‡}

¹*School of Mathematical Sciences, Shanghai Jiao Tong University, Shanghai 200240, China*

²*Center for Computational Mathematics, Flatiron Institute, Simons Foundation, New York 10010, USA*

³*CMA-Shanghai and MOE-LSC, Shanghai Jiao Tong University, Shanghai 200240, China*

Machine-learning interatomic potentials have emerged as a revolutionary class of force-field models in molecular simulations, delivering quantum-mechanical accuracy at a fraction of the computational cost and enabling the simulation of large-scale systems over extended timescales. However, they often focus on modeling local environments, neglecting crucial long-range interactions. We propose a Sum-of-Gaussians Neural Network (SOG-Net), a lightweight and versatile framework for integrating long-range interactions into machine learning force field. The SOG-Net employs a latent-variable learning network that seamlessly bridges short-range and long-range components, coupled with an efficient Fourier convolution layer that incorporates long-range effects. By learning sum-of-Gaussian multipliers across different convolution layers, the SOG-Net adaptively captures diverse long-range decay behaviors while maintaining close-to-linear computational complexity during training and simulation via non-uniform fast Fourier transforms. The method is demonstrated effective for a broad range of long-range systems.

Machine-learning interatomic potentials (MLIPs) have become essential tools in atomic force field modeling, bridging the gap between the high accuracy of computationally expensive quantum mechanical calculations and the efficiency of empirical force fields [1–3]. Most MLIPs assume that the potential energy surface of an atom is determined by its local environment within a finite cut-off radius, an assumption rooted in the “nearsightedness principle” of electronic structure theory [4, 5]. This approximation simplifies the model, enabling predictions to scale linearly with system size. However, it systematically neglects long-range (LR) interactions such as Coulomb, dispersion, and dipole forces, which are crucial for accurately modeling the heterogeneous structures and functionalities of polar materials and biological systems in molecular dynamics (MD) simulations [6, 7]. Incorporating LR interactions into MLIP models while preserving efficiency and scalability remains a significant challenge in the field.

To capture LR effects in MLIP models, most efforts focused on electrostatic interactions. Early approaches [8, 9] assigned fixed point charges with empirical force-field corrections; however, these methods struggled with charge-transfer effects and the challenge of defining accurate corrections. Recent advancements include the use of virtual charge sites, as employed in deep potential LR [10] and self-consistent field neural networks [11] as well as methods that predict effective partial charges while enforcing global charge conservation [12–14]. In these models, once latent variables such as virtual sites or partial charges are determined, electrostatic interactions are computed using classical algorithms such as Ewald summation [15] or self-consistent field iterations. These approaches have demonstrated success in applications

such as phase diagram predictions [16, 17]. However, accurately and efficiently modeling diverse LR decay behaviors in MLIPs remains far from settled. The classical Ewald summation [15], commonly used to account for LR effects, was originally designed for $1/r$ Coulomb potentials and is ill-suited for other LR decay rates, such as $1/r^p$ (with $p > 1$ or non-integer), $e^{-\mu r}/r$ (with $\mu > 0$), or more complex forms involving combinations of decay exponents [18]. Approaches such as the long-distance equivariant (LODE) method [19, 20] improve the accuracy for inverse power-law decays but rely heavily on the accuracy of the initial LR tail guess. Similarly, message passing neural networks [21–25] capture nonlocal interactions through stacked graph convolution layers but involve prohibitive computational costs for slowly decaying LR potentials.

This paper proposes a sum-of-Gaussians neural network (SOG-Net) for accurately learning interatomic potentials with general LR tails. It separates atomic interactions into short-range (SR) and LR components, which are connected through a network that learns latent variables. These variables are processed by an efficient LR convolution layer that accounts for LR contributions, where the SOG multiplier in each layer is represented as a sum of Gaussians with trainable variances and amplitudes. The resulting SOG-Net effectively captures LR effects while maintaining nearly linear computational complexity during both training and simulation through the use of non-uniform fast Fourier transform (NUFFT) [26, 27]. Unlike existing works relying on either various forms of Ewald summation or self-consistent iterations to account for LR contributions, our method adaptively captures diverse LR decay tails present in the system by learning Gaussian multipliers across different convolution layers.

Consider a system of N atoms at $\mathbf{r}_i$, for $i = 1, \dots, N$, in a three-dimensional space with periodic boundary conditions. One assumes that the total potential energy can

* jiyajie595@sjtu.edu.cn

† Contact author: jliang@flatironinstitute.org

‡ Contact author: xuzl@sjtu.edu.cn

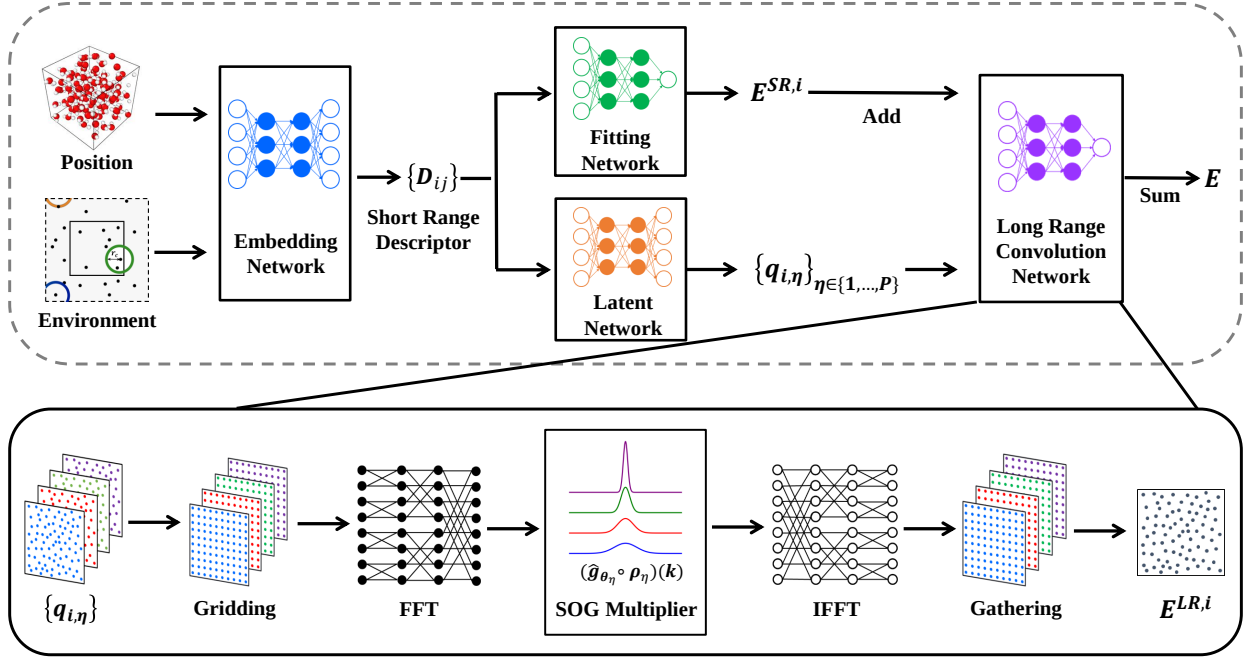


FIG. 1. Schematic illustration of the SOG-Net model. The upper section depicts the entire network structure, whereas the lower section provides a detailed view of the LR convolution layer. Here, $\{q_{i,\eta}\}$, $E^{SR,i}$, and $E^{LR,i}$ denote the latent variables and SR and LR component of atomic energy, respectively.

be written into SR and LR parts,

$$E = E^{SR} + E^{LR}. \quad (1)$$

As is standard in most MLIPs [28], the SR energy is summed over the atomic contribution of each atom:

$$E^{SR} = \sum_{i=1}^N E^{SR,i} = \sum_{i=1}^N f_{\theta_{SR}}(\mathbf{D}_i), \quad (2)$$

where $\mathbf{D}_i$ is the SR descriptor, comprising invariant features of the i -th atom. The neural network $f_{\theta_{SR}}(\cdot)$, parameterized by θ_{SR} , maps $\mathbf{D}_i$ to the atomic SR energy $E^{SR,i}$. The descriptor $\mathbf{D}_i$ is a function of the local atomic environment $\mathcal{R}_i \in \mathbb{R}^{N_i \times 3}$, where N_i represents the number of neighbors of atom i within a predefined cutoff r_c . To ensure physical consistency, $\mathbf{D}_i$ preserves symmetries such as translation, rotation, and permutation. Further details on its practical construction are provided in the Supplementary Materials (SM), Section II [29].

Leveraging the rapid decay of LR interactions in Fourier space, we model the LR energy using multiple Fourier convolution layers:

$$E^{LR} = \frac{1}{2V} \sum_{i=1}^N \sum_{\eta=1}^P q_{i,\eta} \sum_{\mathbf{k}} \hat{g}_{\theta_\eta}(\mathbf{k}) \hat{\rho}_\eta(\mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{r}_i}, \quad (3)$$

where $P \in \mathbb{Z}^+$ is the total number of layers,

$$\mathbf{q}_\eta = (q_{1,\eta}, \dots, q_{N,\eta}), \quad \eta = 1, \dots, P \quad (4)$$

denotes the η -th layer of latent variables for the LR energy, $\hat{g}_{\theta_\eta}$ is the SOG multiplier operator parameterized by θ_η for the η -th layer, $\mathbf{k} = 2\pi(m_x/L_x, m_y/L_y, m_z/L_z)$ denotes the Fourier modes with $m_x, m_y, m_z \in \mathbb{Z}$, and

$$\hat{\rho}_\eta(\mathbf{k}) = \sum_{j=1}^N q_{j,\eta} e^{-i\mathbf{k} \cdot \mathbf{r}_j} \quad (5)$$

denotes the structure factor of the latent variables for the η -th layer. In the LR model, the latent variables $\mathbf{q}_\eta$ in each layer can be interpreted as the weights assigned to each atom for different types of LR interactions, such as partial charges in electrostatic interactions or atomic sizes in dispersion interactions. These latent variables are obtained by mapping the invariant features $\mathbf{D}_i$ to $\mathbf{q}_\eta$ using a latent neural network parameterized by $\theta_{l_{nn}}$, i.e., $\{q_{i,\eta}\}_{\eta=1}^P = f_{\theta_{l_{nn}}}(\mathbf{D}_i)$. A schematic diagram of the basic network structure is presented in the upper half of Figure 1.

In our LR model, two key aspects are the parameterization of the SOG multiplier operator $\hat{g}_{\theta_\eta}$ and the efficiency in computing Eq. (3). These factors will be addressed sequentially as follows. We parameterize $\hat{g}_{\theta_\eta}$ as a sum of Gaussians,

$$\hat{g}_{\theta_\eta}(\mathbf{k}) = \sum_{\ell=0}^M w_{\ell,\eta} e^{-k^2/s_{\ell,\eta}^2}, \quad (6)$$

where $\theta_\eta = \{w_{\ell,\eta}, s_{\ell,\eta}\}_{\ell=0}^M$ is the set of trainable parameters. The SOG multipliers $\{\hat{g}_{\theta_\eta}\}_{\eta=1}^P$ serve as a multilayer approximation of the LR tail of the potential in

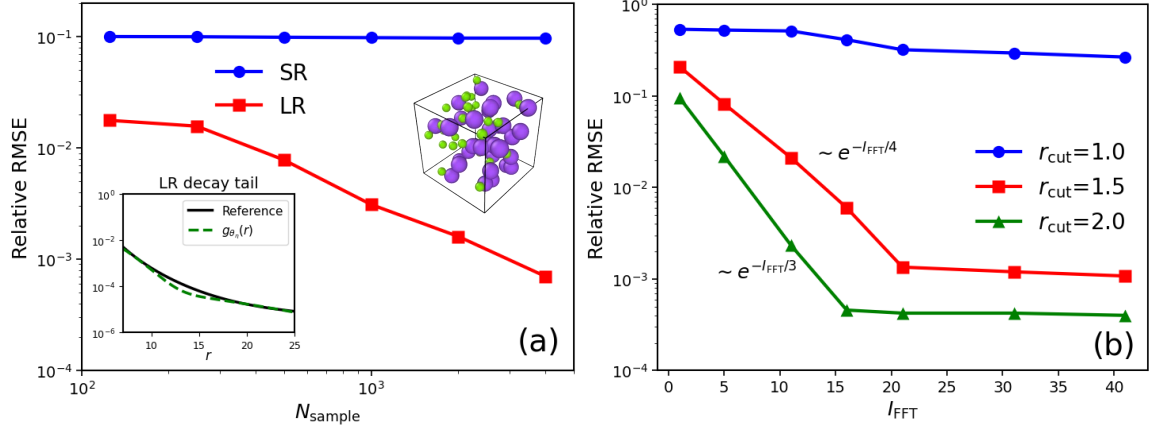


FIG. 2. (a) Plot of the test errors as a function of the size of training set. Data are shown for the short-range component (SR) and the full SOG-Net model (LR). The insets at the upper right and bottom left display a snapshot of the system and a plot of the learned SOG multiplier g_{θ_η} , respectively. The solid line represents the reference LR decaying tail in the force field. (b) Plot of the test error, represented by the relative root mean square error (RMSE), as a function of the number of Fourier grids l_{FFT} . The data are shown for different real-space cutoff radii r_c .

Fourier space. Gaussians exhibit excellent symmetry and smoothness and are renowned for minimizing the Heisenberg uncertainty principle for L^2 functions [30]. These properties allow the SOG multipliers to achieve an optimal trade-off between spatial and frequency localization, facilitating the effective representation of LR tails. The form of LR energy, as constructed in Eqs. (3)-(6), inherently preserves translation, rotation, and permutation invariants. Additional constraints can be imposed either on the coefficients θ_η or by projecting the learned potential onto a set of symmetric basis functions [19, 31].

In practice, the weights $\{w_{\ell,\eta}\}_{\ell=0}^M$ in the SOG are initialized as a vector of ones for $\eta = 1, \dots, P$, and the bandwidths are initialized as

$$s_{\ell,\eta} = e^{b_{\min} + \ell(b_{\max} - b_{\min})/M}, \quad \ell = 0, \dots, M, \quad (7)$$

where $b_{\min}, b_{\max} \in \mathbb{R}$, and $b_{\min} < b_{\max}$. This logarithmically spacing generates $s_{\ell,\eta}$ as points in the range $[e^{b_{\min}}, e^{b_{\max}}]$, effectively capturing the multiscale nature of LR potentials and significantly reducing the number of Gaussians needed for accurate representation. Additionally, after completing the training, M is further reduced using model reduction techniques [32] to improve the prediction efficiency. Further details on the parameter initialization and model reduction method are provided in the SM (Section I C and I D) [29].

Here, we further propose a fast algorithm to accelerate our model, achieving near-optimal complexity and improved scalability. The method begins by mapping each layer of latent variables, $q_{i,\eta}$, onto a uniform grid through convolution with a shape function W , referred to as the gridding operator $\mathcal{G}_\eta$. The FFT operator $\mathcal{F}$ is then applied to compute $\hat{\rho}_\eta(\mathbf{k})$. For each layer, the diagonal scaling operator $\mathcal{S}_\eta$ performs pointwise multiplication $(\hat{g}_{\theta_\eta} \circ \hat{\rho}_\eta)(\mathbf{k})$ in the Fourier space. Subsequently, the inverse FFT operator $\mathcal{F}^{-1}$ recovers the atomic LR

energy at each grid point. Finally, the gathering operator $\mathcal{G}_\eta^{-1}$ maps the atomic energy back to the atomic positions $\{\mathbf{r}_i\}_{i=1}^N$, summing contributions across all layers. It is worth noting that the gridding, scaling, and gathering operators can be applied species-wise in practical implementations. The entire process of the LR convolution layer is summarized as

$$E^{LR,i} = \mathcal{G}_\eta^{-1} \mathcal{F}^{-1} \mathcal{S}_\eta \mathcal{F} \mathcal{G}_\eta \left[\{q_{i,\eta}\}_{\eta=1}^P \right], \quad (8)$$

with a schematic illustration provided in the lower half of Figure 1. The detailed method is provided in the SM, Section III [29], where we leverage the FINUFFT package [33] to accelerate these operators via NUFFTs. As a result, each operator in Eq. (8) can be executed in a complexity of either $O(N + N_{\text{FFT}} \log N_{\text{FFT}})$, where N_{FFT} represents the average number of grid points per layer. This highlights a key advantage of the SOG-Net: after the learning process is complete, the model inherently supports acceleration through a fast algorithm, thereby enhancing the simulation process.

To evaluate the model, we start by applying SOG-Net to a toy model of NaCl electrolytes. This system consists of 1000 particles interacting through Coulomb and dispersion forces. The training set contains at most 4000 configurations, and the test set includes 200 configurations, both sampled every 1000 steps from a long MD trajectory using LAMMPS [34] in the NVT ensemble at $T = 300$ K. Further details on the training setup for the SR component are provided in the SM, Section IV [29]. For the LR component, we use a single layer of latent variables ($P = 1$) and six Gaussians ($M = 6$) in the LR convolution layer. In Figure 2(a), we plot the test error as a function of the training set size, keeping the number of Fourier grids fixed as $N_{\text{FFT}} = 21^3$. The results indicate that using either the SR or LR parts alone results in limitations in the representational capacity. However,

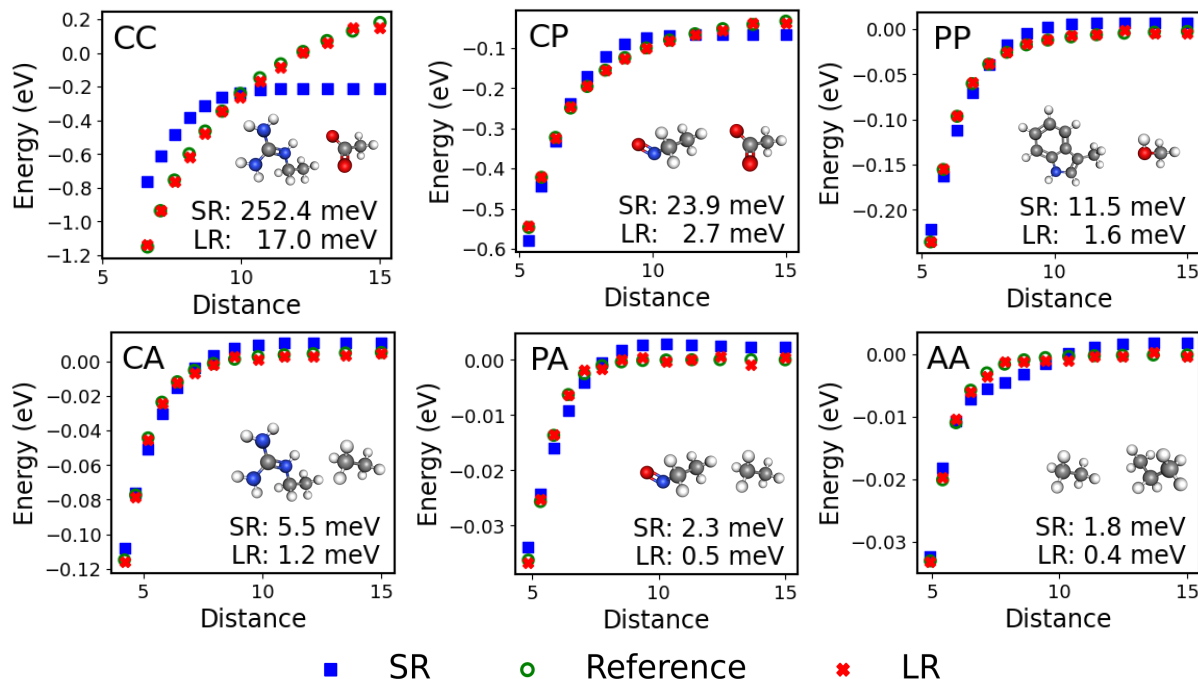


FIG. 3. Comparison of the binding energy curves for the short-range component (SR) and the full SOG-Net model (LR) across six dimer classes: charged-charged (CC), charged-polar (CP), polar-polar (PP), charged-apolar (CA), polar-apolar (PA), and apolar-apolar (AA). The binding energy curve represents the potential energy difference between the dimer and its isolated monomers. Insets display system snapshots with charge states labeled, along with the RMSE of energy for both models.

combining the two components improves the accuracy by nearly 1–2 orders of magnitude. The inset on the bottom left of Figure 2(a) demonstrates that our SOG multiplier effectively captures the LR tail, which is represented by a mix of electrostatic and Lennard-Jones interactions. In Figure 2(b), we plot the test error of the complete network against the number of Fourier grids. When the cutoff radius of the SR network exceeds $2^{1/6}$, the equilibrium distance for dispersion interactions, the SOG-Net achieves a relative error of 10^{-3} using only 21 grid points per dimension. Conversely, when the cutoff radius is less than $2^{1/6}$, incorporating an LR layer fails to improve the fitting accuracy due to insufficient representation of repulsive interactions. These findings suggest that the choice of parameters for the SR and LR parts must be carefully balanced to optimize the accuracy.

Next, we evaluate the SOG-Net by examining the binding curves of dimer pairs formed from charged (C), polar (P), and apolar (A) relaxed molecules at various separations (ranging from approximately $5\text{\AA}$ to $15\text{\AA}$) within a periodic cubic box of side length $L = 30\text{\AA}$. These three molecular categories result in six distinct dimer classes (CC, CP, PP, CA, PA, and AA), each characterized by ideal power-law decay constants $1/r^p$ for their interactions: $p_{CC} = 1$, $p_{CP} = 2$, $p_{PP} = 3$, $p_{CA} = 4$, $p_{PA} = 5$, and $p_{AA} = 6$. Snapshots of the dimer configurations are provided as insets in Figure 3. Our benchmark utilizes data from the BioFragment Database [35], computed using the HSE06 hybrid functional [36] with a high fraction

of exact exchange. For each molecular pair, the dataset is randomly split into training and testing sets in a 9 : 1 ratio. This benchmark is designed to evaluate the ability of the SOG-Net model to accurately capture the LR behavior of various interaction types. In the LR component of the SOG-Net, we use a single layer of latent variables, $M = 12$ Gaussians, and $N_{\text{FFT}} = 31^3$ Fourier grids. The construction of the SR component and the training setup are detailed in the SM, Section IV [29]. Figure 3 presents the binding energy curve. The pure SR models fail to accurately capture the training data, as evidenced by the noticeable flattening of the binding energy curves at larger separations between the two molecules. The improvement achieved by our LR models is particularly remarkable for the CA, PA, and AA cases, where the energy scales are much smaller (below 0.1 meV). Moreover, the RMSE of forces are presented in Figure SI in the SM [29]. The results demonstrate that the model incorporating the LR component significantly outperforms the pure SR model in all cases, with the RMSE of the forces reduced by approximately an order of magnitude. These observations highlight the critical role of the LR component in accurately modeling atomic interactions.

Below, we highlight the advantages of the SOG-Net compared to previous methods tested on the same dataset. In the generalized LODE method [19], the training accuracy depends heavily on guessing the potential exponent p for the LR tail of each dimer class. In contrast, the SOG-Net eliminates the need for such manual

tuning, as its SOG multiplier adaptively fits the LR tail during training. Additionally, the recently proposed latent Ewald method [37] also learns latent variables from invariant features but employs the Ewald summation for evaluating the LR energy. The latent Ewald approach is well-suited for interactions with a $1/r$ LR tail, offering benefits for the CC class; however, it yields only modest improvements for non-CC classes. The SOG-Net is able to accurately capture diverse LR tails across both CC and five non-CC cases. For example, in the CC, CP, and PP classes, the force RMSEs are 4.5, 4.1, and 0.51 meV, respectively, compared to approximately 18, 20, and 5 meV for the LODE method (with optimal p) and 5.9, 16.5, and 3.0 meV for the latent Ewald approach. These results demonstrate the outstanding performance of the SOG-Net.

Finally, we applied the SOG-Net to a dataset of 1900 liquid-water configurations, each containing 300 atoms. The system has a density of 1 g/ml at $T = 300$ K. The dataset was generated via DFT calculations using the projector-augmented wave method [38]. The electronic exchange-correlation energy was described using the generalized gradient approximation with the Perdew–Burke–Ernzerhof functional [39]. Details of the training setup for the SR component are provided in the SM, Section IV [29]. For the LR component, we use $P = 16$ layers of latent variables, $M = 12$ Gaussians for the SOG multiplier, and 31 Fourier modes for each direction. The learning curves of SOG-Net, shown in Figure SII in the SM [29], demonstrate that incorporating the LR component into the SOG-Net significantly reduces the error compared with the pure SR component.

We performed MD simulations under the NVT ensemble. Figure 4(a) displays the oxygen-oxygen radial distribution function (RDF). RDFs for the oxygen-hydrogen and hydrogen-hydrogen pairs are provided in the SM (Figure SIII) [29]. All the computed RDFs are nearly identical and show excellent agreement with the DFT results. In Figure 4(b), we present the longitudinal component of the charge structure factor along the z -axis, defined as

$$S(\mathbf{k}) = \frac{1}{V} \sum_{ij=1}^N q_i q_j \langle e^{i\mathbf{k} \cdot \mathbf{r}_{ij}} \rangle, \quad (9)$$

where $\langle \cdot \rangle$ denotes the ensemble average. The charge structure factor in the long-wavelength limit reflects the accuracy of reproducing LR correlations in water [40]. The results show that our LR model could adequately describe the charge structure factors, whereas pure SR models diverge sharply as $k \rightarrow 0$, consistent with observa-

tions in prior studies on SR and LR electrostatics [41, 42]. Additionally, we compute the average number of hydrogen bonds per water molecule using the pure SR model and the full SOG-Net model, as shown in the inset of Figure 4(b). The results confirm that incorporating the LR component significantly enhances the accuracy. The time performance of the SOG-Net during MD simulations is given in Figure SIV of the SM [29], demonstrating the linear scaling of the model. Although the theoretical complexity of the FFT and IFFT operators is $O(N \log N)$, their contribution to the overall computation time in the LR convolution layer is minimal. The CPU time of the full SOG-Net model is only slightly higher (10% – 40%, depending on the system size) than that of the pure SR model, indicating that SOG-Net achieves significant improvements in fitting accuracy with only a modest increase in computational cost.

In conclusion, we have developed an efficient and versatile framework, SOG-Net, for modeling LR interactions in atomistic systems. Unlike existing LR methods, the SOG-Net requires no user-defined electrostatics or dispersion baseline corrections and does not rely on the classical Ewald summation, which assumes a predefined decay rate of $1/r$. It exhibits spectral convergence in learning and testing errors with the number of Fourier modes. Moreover, the LR MLIP incurs only a modest computational overhead compared to the pure SR network. This cost can be further reduced for both SR and LR components by leveraging acceleration techniques, such as random batch-type schemes [43, 44]. By addressing these challenges, the SOG-Net provides a robust foundation for large-scale simulations with first-principles accuracy that account for LR interactions. Notably, the LR convolution layer in the SOG-Net can also be integrated into existing methods such as HDNNP [45], ACE [46], MACE [23], CACE [47], GAP [8], and MTPs [48]. In these approaches, after latent variables, such as partial charges, Wannier centers, or electronegativity, are learned through existing frameworks [10–13, 49], the SOG-Net can be readily integrated to enable adaptive learning of LR decay tails. This integration provides a simple mathematical formulation and achieves near-optimal computational complexity.

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[1] P. Friederich, F. Häse, J. Proppe, and A. Aspuru-Guzik, Machine-learned potentials for next-generation matter

simulations, *Nat. Mater.* **20**, 750 (2021).

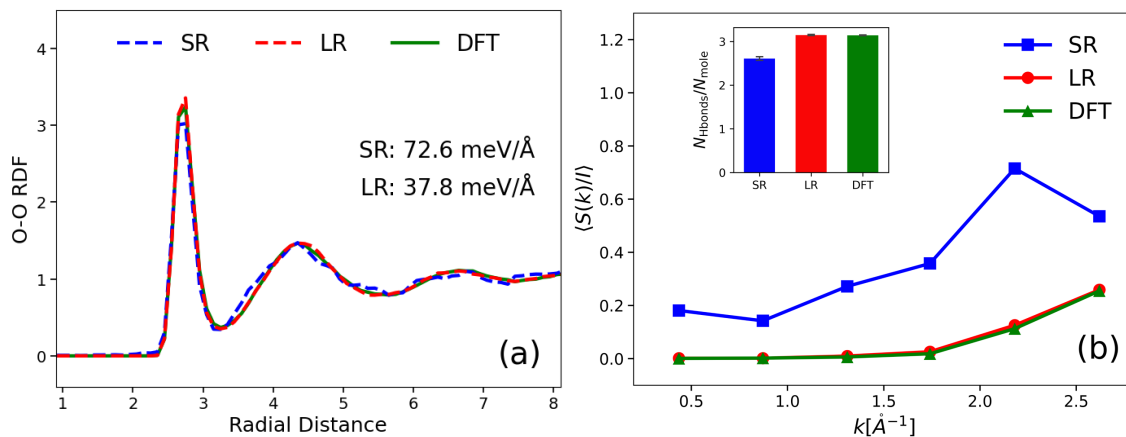


FIG. 4. (a) Predicted radial distribution functions (RDFs) of oxygen-oxygen (O-O) atom pairs of water at $T = 300$ K and 1g/mL , using SR or LR models compared with the DFT results. The inset provides a snapshot of the simulated system, and the test errors are listed in the lower right corner. (b) Predicted charge-charge structure correlations over charge density in reciprocal space as a function of k values, using SR or LR models. The inset presents a plot of the average number of hydrogen bonds per water molecule.

- [2] O. T. Unke, S. Chmiela, H. E. Sauceda, M. Gastegger, I. Poltavsky, K. T. Schütt, A. Tkatchenko, and K. R. Müller, Machine learning force fields, *Chem. Rev.* **121**, 10142 (2021).
- [3] L. Zhang, J. Han, H. Wang, R. Car, and W. E, Deep potential molecular dynamics: a scalable model with the accuracy of quantum mechanics, *Phys. Rev. Lett.* **120**, 143001 (2018).
- [4] W. Kohn, Density functional and density matrix method scaling linearly with the number of atoms, *Phys. Rev. Lett.* **76**, 3168 (1996).
- [5] E. Prodan and W. Kohn, Nearsightedness of electronic matter, *P. Nat. Acad. Sci.* **102**, 11635 (2005).
- [6] H.-X. Zhou and X. Pang, Electrostatic interactions in protein structure, folding, binding, and condensation, *Chem. Rev.* **118**, 1691 (2018).
- [7] J. Yuan and H. Tanaka, Charge regulation effects in polyelectrolyte adsorption, *Phys. Rev. Lett.* **132**, 228101 (2024).
- [8] A. P. Bartók, M. C. Payne, R. Kondor, and G. Csányi, Gaussian approximation potentials: The accuracy of quantum mechanics, without the electrons, *Phys. Rev. Lett.* **104**, 136403 (2010).
- [9] O. T. Unke, S. Chmiela, M. Gastegger, K. T. Schütt, H. E. Sauceda, and K. R. Müller, SpookyNet: Learning force fields with electronic degrees of freedom and nonlocal effects, *Nat. Commun.* **12**, 7273 (2021).
- [10] L. Zhang, H. Wang, M. C. Muniz, A. Z. Panagiotopoulos, R. Car, *et al.*, A deep potential model with long-range electrostatic interactions, *J. Chem. Phys.* **156**, 124107 (2022).
- [11] A. Gao and R. C. Remsing, Self-consistent determination of long-range electrostatics in neural network potentials, *Nat. Commun.* **13**, 1572 (2022).
- [12] O. T. Unke and M. Muwly, PhysNet: A neural network for predicting energies, forces, dipole moments, and partial charges, *J. Chem. Theory Comput.* **15**, 3678 (2019).
- [13] T. W. Ko, J. A. Finkler, S. Goedecker, and J. Behler, A fourth-generation high-dimensional neural network potential with accurate electrostatics including non-local charge transfer, *Nat. Commun.* **12**, 398 (2021).
- [14] Y. Shaidu, F. Pellegrini, E. Küçükbenli, R. Lot, and S. de Gironcoli, Incorporating long-range electrostatics in neural network potentials via variational charge equilibration from shortsighted ingredients, *NPJ Comput. Mater.* **10**, 47 (2024).
- [15] P. P. Ewald, Die Berechnung optischer und elektrostatischer Gitterpotentiale, *Ann. Phys.* **369**, 253 (1921).
- [16] L. Zhang, H. Wang, R. Car, and W. E, Phase diagram of a deep potential water model, *Phys. Rev. Lett.* **126**, 236001 (2021).
- [17] S. L. Bore and F. Paesani, Realistic phase diagram of water from “first principles” data-driven quantum simulations, *Nat. Commun.* **14**, 3349 (2023).
- [18] J. N. Israelachvili, *Intermolecular and Surface Forces* (Academic Press, 2011).
- [19] K. K. Huguenin-Dumittan, P. Loche, N. Haoran, and M. Ceriotti, Physics-inspired equivariant descriptors of nonbonded interactions, *J. Phys. Chem. Lett.* **14**, 9612 (2023).
- [20] A. Grisafi and M. Ceriotti, Incorporating long-range physics in atomic-scale machine learning, *J. Chem. Phys.* **151**, 204105 (2019).
- [21] K. Schütt, P.-J. Kindermans, H. E. Sauceda Felix, S. Chmiela, A. Tkatchenko, and K. R. Müller, Schnet: A continuous-filter convolutional neural network for modeling quantum interactions, *Adv. Neural Inf. Process. Syst. (NeurIPS)* **30**, 992 (2017).
- [22] S. Batzner, A. Musaelian, L. Sun, M. Geiger, J. P. Mailoa, M. Kornbluth, N. Molinari, T. E. Smidt, and B. Kozinsky, E(3)-equivariant graph neural networks for data-efficient and accurate interatomic potentials, *Nat. Commun.* **13**, 2453 (2022).
- [23] I. Batatia, D. P. Kovacs, G. Simm, C. Ortner, and G. Csányi, MACE: Higher order equivariant message passing neural networks for fast and accurate force fields, *Adv. Neural Inf. Process. Syst. (NeurIPS)* **35**, 11423 (2022).
- [24] A. Kosmala, J. Gastegger, N. Gao, and S. Günnemann, Ewald-based long-range message passing for molecular

- graphs, in *International Conference on Machine Learning* (ICML, 2023) pp. 17544–17563.
- [25] B. Deng, P. Zhong, K. Jun, J. Riebesell, K. Han, C. J. Bartel, and G. Ceder, CHGNet as a pretrained universal neural network potential for charge-informed atomistic modelling, *Nat. Mach. Intell.* **5**, 1031 (2023).
 - [26] A. Dutt and V. Rokhlin, Fast Fourier transforms for nonequispaced data, *SIAM J. Sci. Comput.* **14**, 1368 (1993).
 - [27] L. Greengard and J.-Y. Lee, Accelerating the nonuniform fast Fourier transform, *SIAM Rev.* **46**, 443 (2004).
 - [28] T. W. Ko and S. P. Ong, Recent advances and outstanding challenges for machine learning interatomic potentials, *Nat. Comput. Sci.* **3**, 998 (2023).
 - [29] See supplementary materials for detailed mathematical derivations, numerical quadrature scheme, its error analysis, and numerical validations, [ur1], includes references [8,10,13,14,19-28,31,32,36,46,47].
 - [30] G. B. Folland and A. Sitaram, The uncertainty principle: a mathematical survey, *J. Fourier Anal. App.* **3**, 207 (1997).
 - [31] C. Faller, M. Kaltak, and G. Kresse, Density-based long-range electrostatic descriptors for machine learning force fields, *J. Chem. Phys.* **161** (2024).
 - [32] P. Benner, S. Gugercin, and K. Willcox, A survey of projection-based model reduction methods for parametric dynamical systems, *SIAM Rev.* **57**, 483 (2015).
 - [33] A. H. Barnett, J. Magland, and L. af Klinteberg, A parallel nonuniform fast Fourier transform library based on an “exponential of semicircle” kernel, *SIAM J. Sci. Comput.* **41**, C479 (2019).
 - [34] A. P. Thompson, H. M. Aktulga, R. Berger, D. S. Bolinteanu, W. M. Brown, P. S. Crozier, P. J. In’t Veld, A. Kohlmeyer, S. G. Moore, T. D. Nguyen, *et al.*, LAMMPS—a flexible simulation tool for particle-based materials modeling at the atomic, meso, and continuum scales, *Comput. Phys. Commun.* **271**, 108171 (2022).
 - [35] L. A. Burns, J. C. Faver, Z. Zheng, M. S. Marshall, D. G. Smith, K. Vanommeslaeghe, A. D. MacKerell, K. M. Merz, and C. D. Sherrill, The BioFragment Database (BFDdb): An open-data platform for computational chemistry analysis of noncovalent interactions, *J. Chem. Phys.* **147**, 161727 (2017).
 - [36] J. Heyd, G. E. Scuseria, and M. Ernzerhof, Hybrid functionals based on a screened Coulomb potential, *J. Chem. Phys.* **118**, 8207 (2003).
 - [37] B. Cheng, Latent Ewald summation for machine learning of long-range interactions, arXiv:2408.15165 (2024).
 - [38] G. Kresse and J. Furthmüller, Efficient iterative schemes for ab initio total-energy calculations using a plane-wave basis set, *Phys. Rev. B* **54**, 11169 (1996).
 - [39] M. C. Payne, M. P. Teter, D. C. Allan, T. Arias, and a. J. Joannopoulos, Iterative minimization techniques for ab initio total-energy calculations: molecular dynamics and conjugate gradients, *Rev. Mod. Phys.* **64**, 1045 (1992).
 - [40] Z. Hu, The symmetry-preserving mean field condition for electrostatic correlations in bulk, *J. Chem. Phys.* **156** (2022).
 - [41] S. J. Cox, Dielectric response with short-ranged electrostatics, *P. Nat. Acad. Sci.* **117**, 19746 (2020).
 - [42] A. Schlaich, E. W. Knapp, and R. R. Netz, Water dielectric effects in planar confinement, *Phys. Rev. Lett.* **117**, 048001 (2016).
 - [43] J. Liang, Z. Xu, and Y. Zhao, Random-batch list algorithm for short-range molecular dynamics simulations, *J. Chem. Phys.* **155**, 044108 (2021).
 - [44] J. Liang, Z. Xu, and Q. Zhou, Random batch sum-of-Gaussians method for molecular dynamics simulations of particle systems, *SIAM J. Sci. Comput.* **45**, B591 (2023).
 - [45] J. Behler and M. Parrinello, Generalized neural-network representation of high-dimensional potential-energy surfaces, *Phys. Rev. Lett.* **98**, 146401 (2007).
 - [46] R. Drautz, Atomic cluster expansion for accurate and transferable interatomic potentials, *Phys. Rev. B* **99**, 014104 (2019).
 - [47] B. Cheng, Cartesian atomic cluster expansion for machine learning interatomic potentials, *NPJ Comput. Mater.* **10**, 157 (2024).
 - [48] A. V. Shapeev, Moment tensor potentials: A class of systematically improvable interatomic potentials, *Multi-scale Model. Simul.* **14**, 1153 (2016).
 - [49] A. K. Rappe and W. A. Goddard III, Charge equilibration for molecular dynamics simulations, *J. Phys. Chem.* **95**, 3358 (1991).

Supplementary Material of “Machine-Learning Interatomic Potentials for Long-Range Systems”

I. METHODS FOR THE LONG-RANGE NETWORK

In this section, we discuss the motivation and technical details of the long-range (LR) convolution layer used in our sum-of-Gaussians neural network (SOG-Net), including the derivation of the LR model, the scheme for the parameter selection of Gaussian multipliers, and the implementation details of the trained model.

A. Limitations on using the Ewald summation in MLIPs

Our primary motivation for developing the SOG-Net model is to enhance the widely used Ewald summation framework in machine learning interatomic potentials (MLIPs). In this subsection, we first discuss why the Ewald summation framework is unsuitable for LR tails that do not follow the $1/r^p$ form. We then introduce our improved model to address these limitations.

In classical models, charged particles are often modeled as point charges, with pairwise interactions following the empirical Coulomb form $1/r$. Due to the long-range nature, direct computation has an unacceptably high complexity of $O(N^2)$. To accelerate calculations, the Coulomb potential is often decomposed by Ewald splitting into the form

$$\frac{1}{r} = \frac{\text{erfc}(\alpha r)}{r} + \frac{\text{erf}(\alpha r)}{r}, \quad (\text{I.1})$$

where $\text{erf}(\cdot)$ and $\text{erfc}(\cdot)$ represent the error function and complementary error function, respectively, and α is known as the Ewald splitting parameter. The $\text{erfc}(\alpha r)/r$ term decays rapidly in real space and can thus be truncated for efficient computation. The $\text{erf}(\alpha r)/r$ term is singular in real space but very smooth, with

$$\frac{\text{erf}(\alpha r)}{r} \rightarrow \frac{1}{r}, \quad \text{as } r \rightarrow \infty \quad \text{and} \quad \alpha \in (0, \infty), \quad (\text{I.2})$$

making it suitable for truncation in Fourier space. Suppose that N particles, $\{\mathbf{r}_i\}_{i=1}^N$, are located within a cubic box $\Omega = [0, L]^3$ with weights $\{q_i\}_{i=1}^N$. The periodic boundary conditions are imposed. The contributions on potential from the $\text{erf}(\alpha r)/r$ term is given by

$$\Phi(\mathbf{r}_i) = \sum_{\mathbf{n} \in \mathbb{Z}^3} \sum_{j=1}^N \frac{q_j \text{erf}(\alpha |\mathbf{r}_{ij} + L\mathbf{n}|)}{|\mathbf{r}_{ij} + L\mathbf{n}|}, \quad (\text{I.3})$$

where $\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$ and the prime indicates that the case $i = j$ and $\mathbf{n} = 0$ is excluded from the summation. Its Fourier spectral expansion can be expressed as

$$\Phi(\mathbf{r}_i) = \frac{2\pi}{V} \sum_{\mathbf{k} \neq 0} \frac{e^{-|\mathbf{k}|^2/(4\alpha^2)}}{|\mathbf{k}|^2} |\rho(\mathbf{k})|^2, \quad (\text{I.4})$$

where $\mathbf{k} = 2\pi\mathbf{m}/L$, $\mathbf{m} \in \mathbb{Z}^3$, $\rho(\mathbf{k}) := \sum_{j=1}^N q_j e^{i\mathbf{k}\cdot\mathbf{r}_j}$ denotes the charge structure factor, and $V = L^3$ denotes the volume of Ω .

Unlike classical models, molecules in density functional theory (DFT) simulations are not ideal point particles. Discrepancies arise due to spurious interactions between periodic images, particularly for charged fragments, and noise in binding energies at large distances, especially for weak interactions. Consequently, the LR decay rate of charge-charge interactions can exceed the classical $1/r$ Coulomb potential¹. Other LR interactions, such as dispersion and dipole-dipole interactions, may also deviate from their idealized decay rates. These characteristics are likely to be inherited by MLIPs since they are trained on DFT data.

In the field of MLIP, Ewald summation is a commonly used approach for incorporating LR interactions. Two primary strategies have been developed: the first involves learning localized Wannier centers and introducing virtual charge sites, while the second focuses on predicting partial charges for each site while maintaining a fixed number of sites²⁻⁴. Once these ‘‘latent variables’’ are inferred by the neural network, the Fourier components of the Ewald summation formula, Eq. (I.4), are employed to calculate the contribution of charges to the potential energy. Recent graph-based MLIP studies have also utilized the Ewald summation for modeling LR message passing⁵. However, the use of Ewald summation, as described in Eq. (I.4), assumes a predefined LR decay rate of $1/r$, derived from the $\text{erf}(\alpha r)/r$ term in Eq. (I.2). This fixed assumption may lead to systematic errors in MLIPs when the actual LR decay rate deviates from $1/r$.

B. Derivation of long-range model

To address the limitations of existing MLIPs, we replace the Ewald summation with a multilevel LR convolution layer, where the kernel function of each level is learned through a fully connected neural network, enabling adaptive learning of the LR tail while maintaining computational efficiency.

Consider the LR atomic energy in the form of

$$E^{LR,i} = \sum_{\eta=1}^P q_{i,\eta} \sum_{\mathbf{n} \in \mathbb{Z}^d} \sum_{j=1}^N q_{j,\eta} g_{\theta_\eta}(\mathbf{r}_{ij} + L\mathbf{n}), \quad (\text{I.5})$$

where P is the total number of layers, $\{q_{i,\eta} \mid i = 1, \dots, N\}$ are the latent variables of η th layer, and $g_{\theta_\eta}(\cdot)$ is a kernel function parameterized by θ_η and trained via a neural network. The detailed representation of $g_{\theta_\eta}(\cdot)$ is provided in subsection I C. The LR energy is defined as $E^{LR} = \frac{1}{2} \sum_{i=1}^N E^{LR,i}$, where the factor $1/2$ account for double-counting of interaction energy. In this context, g_{θ_η} and $q_{i,\eta}$ at different layers can be interpreted as representations of various LR interactions and their corresponding coefficients, respectively. For $P = 1$, $q_{i,1}$ is analogous to environment-dependent partial charges on individual atoms. For $P > 1$, the model can incorporate more general LR interactions, thereby enhancing the representation and expressiveness of LR effects in the MLIP.

Let us define the Fourier transform pair as

$$\hat{f}(\mathbf{k}) = \int_{\Omega} f(\mathbf{r}) e^{-i\mathbf{k}\cdot\mathbf{r}} d\mathbf{r}, \quad \text{and} \quad f(\mathbf{r}) = \frac{1}{V} \sum_{\mathbf{k}} \hat{f}(\mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{r}}. \quad (\text{I.6})$$

For a given function f defined on $\mathbb{R}^3$, the Poisson summation formula states that

$$\sum_{\mathbf{n} \in \mathbb{Z}^d} f(\mathbf{r} + L\mathbf{n}) = \frac{1}{V} \sum_{\mathbf{k}} \hat{f}(\mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{r}}. \quad (\text{I.7})$$

By substituting the Poisson summation formula into Eq. (I.5), we derive the Fourier spectral expansion of $E^{LR,i}$:

$$E^{LR,i} = \frac{1}{V} \sum_{\eta=1}^P q_{i,\eta} \sum_{j=1}^N q_{j,\eta} \sum_{\mathbf{k}} \hat{g}_{\boldsymbol{\theta}_\eta}(\mathbf{k}) e^{i\mathbf{k} \cdot (\mathbf{r}_i - \mathbf{r}_j)} - E^{\text{self},i}, \quad (\text{I.8})$$

where

$$E^{\text{self},i} := \frac{1}{V} \sum_{\eta=1}^P |q_{i,\eta}|^2 \sum_{\mathbf{k}} \hat{g}_{\boldsymbol{\theta}_\eta}(\mathbf{k}) \quad (\text{I.9})$$

denotes the unwanted self-interaction energy.

Given the slowly varying nature of LR interactions, we model $g_{\boldsymbol{\theta}_\eta}(\mathbf{r})$ as a radially symmetric function, i.e., $g_{\boldsymbol{\theta}_\eta}(\mathbf{r}) = g_{\boldsymbol{\theta}_\eta}(r)$ with $r = |\mathbf{r}|$. Consequently, its Fourier transform, $\hat{g}_{\boldsymbol{\theta}_\eta}(\mathbf{k}) = \hat{g}_{\boldsymbol{\theta}_\eta}(k)$, is also radially symmetric. We refer to $\hat{g}_{\boldsymbol{\theta}_\eta}(k)$ as the ‘‘SOG multiplier’’, with the parameter $\boldsymbol{\theta}_\eta$ being trainable. Our goal is to develop an efficient and scalable descriptor for the LR component of MLIPs through the following two steps: (1) The short-range (SR) descriptor is fed into a fully connected neural network to generate the latent variables $\{q_{i,\eta}\}_{i=1}^N$ for each layer $\eta = 1, \dots, P$. (2) These latent variables are substituted into Eq. (I.8) to compute $\Phi(\mathbf{r}_i)$, performed using Fourier diagonal scaling with the multiplier $\hat{g}_{\boldsymbol{\theta}_\eta}(k)$. While directly evaluating $\Phi(\mathbf{r}_i)$ for all $i = 1, \dots, N$ requires $O(NN_{\text{tot}})$ operations with N_{tot} the total number of Fourier grids, the LR convolution layer (detailed in Section III) reduces this computation to near-linear complexity in both N and N_{tot} .

C. Parameterization of the SOG multiplier

In this work, we parameterize the SOG multiplier $\hat{g}_{\boldsymbol{\theta}_\eta}$ as

$$\hat{g}_{\boldsymbol{\theta}_\eta}(\mathbf{k}) = \sum_{\ell=0}^M w_{\ell,\eta} e^{-k^2/s_{\ell,\eta}^2}, \quad (\text{I.10})$$

where $\boldsymbol{\theta}_\eta = \{w_{\ell,\eta}, s_{\ell,\eta}\}_{\ell=0}^M$ is the set of trainable parameters of the η th layer. We initialize the weights $\{w_{\ell,\eta}\}$ as matrices of all ones. The bandwidths are initialized as

$$s_{\ell,\eta} = e^{b_{\min}^\eta + \ell(b_{\max}^\eta - b_{\min}^\eta)/M}, \quad \ell = 0, \dots, M, \quad (\text{I.11})$$

where $b_{\min}^\eta, b_{\max}^\eta \in \mathbb{R}$, $b_{\min}^\eta < b_{\max}^\eta$ so that $s_{\ell,\eta}$ are *logarithmically* equally spaced points lying on $[e^{b_{\min}^\eta}, e^{b_{\max}^\eta}]$ for each η . Here, we use logarithmic spacing nodes instead of uniform spacing to capture the multiscale characteristics of LR potentials and reduce the number of Gaussians needed for representation. The periodicity leads to a simple rule to set $b_{\min}^\eta$ apriori for a given error tolerance $\varepsilon > 0$:

$$b_{\min}^\eta \geq \log(2\pi) - \log L - \log \sqrt{\log \varepsilon^{-1}}, \quad (\text{I.12})$$

where this formula ensures that the Gaussian with the smallest bandwidth in the SOG multiplier has its support almost exclusively at the zero mode in Fourier space. Assume that a cutoff radius

r_c is specified for the local environment while constructing the SR descriptor, we also have a simple formula to set $b_{\max}^\eta$:

$$b_{\max}^\eta \leq \log(2r_c^{-1}) + \log \log \varepsilon^{-1} \quad (\text{I.13})$$

so that $g_{\theta_\ell}(\mathbf{r}) \sim \varepsilon$ when $r = r_c$. This ensure that the Gaussian with the largest bandwidth in g_{θ_ℓ} has its support larger than r_c in real space.

The advantages of using an SOG to represent the SOG multiplier are threefold. First, as a type of radial basis function, Gaussians are well-suited for approximating unknown symmetric and smooth LR potentials. Second, Gaussians are renowned for minimizing the Heisenberg uncertainty principle for L^2 functions⁶, achieving an optimal trade-off between spatial and frequency localization. Given that LR potentials frequently exhibit multi-scale characteristics, meaning they have significant support in both real and Fourier spaces, the properties of Gaussian functions make them particularly well-suited for fitting such functions.

The multiplier in Eq. (I.10) is highly versatile and enables the imposition of specific constraints on the learned potential. In our construction, both the multiplier $\hat{g}_{\theta_\eta}$ and its inverse Fourier transform g_{θ_η} are radially symmetric, inherently ensuring translation, inversion, and rotational invariance. Additional constraints can be imposed on the coefficients θ_η by either solving an equation for some certain coefficients after training the others or projecting the learned LR atomic energy $E^{LR,i}$ onto a set of symmetric basis functions defined within the local environment of each atom similar to some previous works^{1,7}.

In comparison to the Gaussian Approximation Potential (GAP)⁸, introduced in early machine learning models, our work presents two key distinctions. First, GAP is primarily designed to fit SR interactions, while our SOG-Net is employed to learn LR interactions in Fourier space. Second, GAP determines the coefficients through Gaussian regression, whereas our method learns θ_η via a neural network.

D. Model Reduction

In practice, a key problem is how to determine the number of Gaussians M in the SOG multiplier. At the start of training, we choose a large M to ensure sufficient resolution. Following each training round, model reduction (MR) techniques⁹ are used to find the optimal number of Gaussians. The MR process starts by converting the multiplier of each layer into a sum of exponentials via the change of variable $K = k^2$ and applying the Laplace transform $\mathcal{L}[\cdot]$, leading to a sum-of-poles representation:

$$\mathcal{L}[\hat{g}_{\theta_\eta}(K)] = \mathcal{L}\left[\sum_{\ell=0}^M w_{\ell,\eta} e^{-K/s_{\ell,\eta}^2}\right] = \sum_{\ell=0}^M \frac{w_{\ell,\eta}}{z + 1/s_{\ell,\eta}^2} = \mathbf{c}(z\mathbf{I} - \mathbf{A})^{-1}\mathbf{b}, \quad (\text{I.14})$$

where $\mathbf{A}$ is an $(M+1) \times (M+1)$ diagonal matrix, $\mathbf{b}$ and $\mathbf{c}$ are column and row vectors of dimension $M+1$, respectively. Next, one regards the rightmost term in Eq. (I.14) as a transfer function of a linear dynamical system, enabling the use of the balanced truncation method to reduce the number of terms to $\widetilde{M} < M+1$. One can refer to the review⁹ for more details.

II. METHODS FOR THE SHORT-RANGE NETWORK

In this section, we discuss the construction of SR component of our SOG-Net as well as some implementation details which is useful for improve the efficiency.

The first step in our model is to generate a robust SR descriptor that preserves symmetry. This descriptor is then fed into two networks: the first, referred to as the “fitting network”, generates the SR potential; the second, called the “latent network”, learns the latent variables $\{q_{i,\eta}\}$ from the SR descriptor. Notably, SOG-Net can be seamlessly integrated with any existing methods for constructing SR descriptors, such as tensor operations in Deep Potential¹⁰ and moment tensor networks¹¹, or basis projections used in MACE¹², CACE¹³, and LODE¹⁴. In the examples presented in this paper, we employ SR descriptors based on tensor operations similar to those in Deep Potential, due to their simplicity of implementation. Exploring alternative SR descriptors and identifying the most suitable approach will be a focus of future work.

Next, we briefly review the construction of the SR model. The SR component of energy, E^{SR} , is computed as the sum of atomic contributions from all atoms:

$$E^{SR} = \sum_{i=1}^N E^{SR,i} = \sum_{i=1}^N f_{\theta_{fit}}(\mathbf{D}_i), \quad (\text{II.1})$$

where $f_{\theta_{fit}}$ is a residual neural network (ResNet), referred to as the “fitting network”, and $\mathbf{D}_i$ is a column vector that represents the local environment of atom i . The vector $\mathbf{D}_i$ is core-shelly structured as $\mathbf{D}_i = \{\mathbf{D}_{i,c}, \mathbf{D}_{i,s}\}$, where $\mathbf{D}_{i,c}$ and $\mathbf{D}_{i,s}$ contain information about neighboring atoms within specified cutoff radii r_c and r_s , respectively, with $r_c < r_s$. The details for constructing $\mathbf{D}_{i,c}$ and $\mathbf{D}_{i,s}$ are provided below. Please note that we may use only $\mathbf{D}_{i,c}$ or $\mathbf{D}_{i,s}$ in numerical tests, depending on the actual situation, as set up in Section IV. The SR component of force, $\mathbf{F}^{SR,i}$, can be derived by applying the automatic differentiation on E^{SR} .

The core component of the SR descriptor, $\mathbf{D}_{i,c}$, is derived from the environment matrix $\mathbf{R}_{i,c}$ and embedding matrix $\mathbf{G}_{i,c}$:

$$\mathbf{D}_{i,c} = \frac{1}{N_c^2} (\mathbf{G}_{i,c})^T \mathbf{R}_{i,c} (\mathbf{R}_{i,c})^T \mathbf{G}_{i,c}^<, \quad (\text{II.2})$$

where N_c denotes the maximum number of neighbors within the cutoff radius r_c , $\mathbf{G}_{i,c}$ and $\mathbf{R}_{i,c}$ have dimensions $N_c \times M_c$ and $N_c \times 4$, respectively. Each row in $\mathbf{R}_{i,c}$ is a four-dimensional vector:

$$\mathbf{R}_{i,c} = \left\{ \text{soft}(r_{ij}) \times \left[\frac{1}{r_{ij}}, \frac{x_{ij}}{r_{ij}^2}, \frac{y_{ij}}{r_{ij}^2}, \frac{z_{ij}}{r_{ij}^2} \right] \mid j = 1, \dots, N_c \right\}, \quad (\text{II.3})$$

where $r_{ij} = |\mathbf{r}_{ij}|$ with $\mathbf{r}_{ij} := \mathbf{r}_i - \mathbf{r}_j$. Here, the soft function $\text{soft}(\cdot)$ is defined by

$$\text{soft}(r) = \begin{cases} 1, & r < j_c r_c, \\ -6 \left(\frac{r - j_c r_c}{r_c - j_c r_c} \right)^5 + 15 \left(\frac{r - j_c r_c}{r_c - j_c r_c} \right)^4 - 10 \left(\frac{r - j_c r_c}{r_c - j_c r_c} \right)^3 + 1, & j_c r_c \leq r < r_c, \\ 0, & r \geq r_c, \end{cases} \quad (\text{II.4})$$

where $0 < j_c < 1$ is a scaling factor determining the smooth transition region. The function $\text{soft}(r)$ smoothly decays from 1 to 0 within the cutoff range r_c , ensuring that its second derivative

is continuous. This property guarantees continuity in the computed forces, which significantly enhances the long-term stability of MD simulations. The embedding matrix $\mathbf{G}_{i,c}$ is derived from the output of a neural network

$$\mathbf{G}_{i,c} = f_{\theta_{emb,c}} \left(\left\{ \frac{\text{soft}(R_{ij})}{|R_{ij}|} \mid j = 1, \dots, N_c \right\} \right), \quad (\text{II.5})$$

where the fully connected embedding neural network $f_{\theta_{emb,c}}$ maps $\text{soft}(R_{ij})/R_{ij}$ to an M_c dimensional vector. A sub-matrix of $\mathbf{G}_{i,c}$ that includes its first $M_c^<$ columns, with $M_c^< < M_c$, is denoted by $\mathbf{G}_{i,c}^<$. The resulting core component of the SR descriptor from Eq. (II.2), $\mathbf{D}_{i,c}$, has dimension $M_c \times M_c^<$, encapsulates both angular and radial information, and preserves symmetry.

The shell component of the SR descriptor, $\mathbf{D}_{i,s}$, is defined as:

$$\mathbf{D}_{i,s} = \frac{1}{N_s} \sum_{j=1}^{N_s} (\mathbf{G}_{i,s})_j, \quad (\text{II.6})$$

where N_s is the maximum number of neighboring atoms within the cutoff r_s , $\mathbf{G}_{i,s} \in \mathbb{R}^{N_s \times M_s}$ is an embedding matrix, and the summation aggregates the rows of $\sum_j$. For fewer than N_s neighbors, the corresponding matrix terms are zero-padded. The embedding matrix $\mathbf{G}_{i,s}$ comprises M_s nodes from the output layer of a neural network function $f_{\theta_{emb,s}}$ applied to $\text{soft}(R_{ij})$:

$$(\mathbf{G}_{i,s})_j = f_{\theta_{emb,s}}(\text{soft}(R_{ij})), \quad (\text{II.7})$$

which includes only radial neighbor information. The soft function, $\text{soft}(\cdot)$, is defined as in Eq. (II.4). In practice, we set $r_c < r_s$, since three-body interactions are more critical for close neighbors, while radial information alone suffices for distant neighbors.

After generating the core component $\mathbf{D}_{i,c}$ and shell component $\mathbf{D}_{i,s}$, $\mathbf{D}_{i,c}$ is flattened into a vector and concatenated with $\mathbf{D}_{i,s}$ to form the short-range descriptor $\mathbf{D}_i = \{\mathbf{D}_{i,c}, \mathbf{D}_{i,s}\}$. The whole process involves constructing a stencil of bins to identify potential neighbors, binning atoms, and looping through the stencil to assemble the neighbor list $\mathcal{N}_i$ based on the core and shell cutoff radii r_c and r_s . To improve efficiency, we employ the kd -tree algorithm, which uses hierarchical spatial decomposition to perform fast nearest-neighbor searches with linear complexity.

After generating the SR descriptor $\mathbf{D}_i$, it is fed into two separate networks. The first is the fitting network, which computes the SR atomic energy as described in Eq. (II.1); the second is a fully connected “latent” network, $f_{\theta_{ltn}}(\cdot)$, which maps the input SR descriptor to output latent variables. These latent variables are then processed by the LR convolution layer to compute the LR atomic energy. The acceleration of this process is outlined in the next section.

III. FFT-ACCELERATED FAST LONG-RANGE CONVOLUTION

In this section, we detail how the fast Fourier transform (FFT) is used to accelerate calculations for the LR convolution layer, as described in Eq. (I.8), within our SOG-Net model. Although the FFT is a well-established approach, however, the non-uniform distribution of atoms in MD systems prevents the direct application of the FFT. Moreover, unlike the Ewald summation which interaction kernel is fully deterministic, our model employs the trainable kernel $\hat{\mathbf{g}}_{\theta_\eta}$, which evolves dynamically during the training process. Below, we propose a fast method to address these challenges.

Let us start by introduce a localized, separable and periodic shape function $W(\mathbf{r})$ along with its Fourier transform $\widehat{W}(\mathbf{k})$ for spreading particles onto a uniform grid and collecting field values from the grid. By inserting the identity $1 \equiv \widehat{W}(\mathbf{k})|\widehat{W}(\mathbf{k})|^{-2}\widehat{W}(\mathbf{k})$ into Eq. (I.8), one has

$$E^{LR,i} = \frac{1}{V} \sum_{\eta=1}^P q_{i,\eta} \sum_{\mathbf{k}} \widehat{W}(\mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{r}} \frac{\widehat{g}_{\theta_\eta}(\mathbf{k})}{|\widehat{W}(\mathbf{k})|^2} \sum_{j=1}^N q_{j,\eta} \widehat{W}(\mathbf{k}) e^{-i\mathbf{k} \cdot \mathbf{r}_j}. \quad (\text{III.1})$$

We will use the last $\widehat{W}(\mathbf{k})$ for the gridding step, the first $\widehat{W}(\mathbf{k})$ for the gathering step, and $|\widehat{W}(\mathbf{k})|^{-2}$ for the diagonal scaling step, respectively. The details are provided as follows.

The calculation of Eq. (III.1) is performed level-by-level. For each level η , let us define

$$\widehat{\Phi}_{\text{grid},\eta}(\mathbf{k}) := \sum_{j=1}^N q_{j,\eta} \widehat{W}(\mathbf{k}) e^{-i\mathbf{k} \cdot \mathbf{r}_j}, \quad (\text{III.2})$$

which is the Fourier transform of

$$\Phi_{\text{grid},\eta}(\mathbf{r}) = \sum_{j=1}^N q_{j,\eta} W(\mathbf{r} - \mathbf{r}_j) \quad (\text{III.3})$$

by the convolution theorem. The evaluating of $\Phi_{\text{grid},\eta}$ on a uniform grid will be defined as the gridding operator $\mathcal{G}_\eta$. After this step is done, we obtain $\widehat{\Phi}_{\text{grid},\eta}$ from $\Phi_{\text{grid},\eta}$ by applying a 3D FFT operator $\mathcal{F}$. Next, we compute

$$\widehat{\Phi}_{\text{scal},\eta}(\mathbf{k}) := \frac{\widehat{g}_{\theta_\eta}(\mathbf{k})}{|\widehat{W}(\mathbf{k})|^2} \widehat{\Phi}_{\text{grid},\eta}(\mathbf{k}) \quad (\text{III.4})$$

for each Fourier mode which is defined as the diagonal scaling operator $\mathcal{S}_\eta$ (here diagonal means without calculations for cross terms). Next, we obtain $\Phi_{\text{scal},\eta}$ on the real-space grid from $\widehat{\Phi}_{\text{scal},\eta}$ by applying an 3D inverse FFT (IFFT) operator $\mathcal{F}^{-1}$. Note that Eq. (III.4) allow us to write Eq. (III.1) as

$$E^{LR,i} = \sum_{\eta=1}^P q_{i,\eta} \int_{\Omega} \Phi_{\text{scal},\eta}(\mathbf{r}') W(\mathbf{r}_i - \mathbf{r}') d\mathbf{r}' := \sum_{\eta=1}^P q_{i,\eta} \Phi_{i,\eta} \quad (\text{III.5})$$

by applying again the convolution theorem. The gathering operator $\mathcal{G}_\eta^{-1}$ is performed by discretizing Eq. (III.5) on a uniform grid using the trapezoidal rule and summing the contributions from grids for each atom. When the shape function W is smooth and compactly supported, the algorithm achieves spectral convergence as the Fourier grid size increases due to the use of FFT. The entire process of the LR convolution layer is summarized as:

$$E^{LR,i} = \sum_{\eta=1}^P q_{i,\eta} \mathcal{G}_\eta^{-1} \mathcal{F}^{-1} \mathcal{S}_\eta \mathcal{F} \mathcal{G}_\eta [\{q_{j,\eta} \mid j = 1, \dots, N; \eta = 1, \dots, P\}]. \quad (\text{III.6})$$

It is worth noting that the gridding, scaling, and gathering operators can be applied species-wise for multi-species systems. Algorithm 1 outlines the procedure for this fast algorithm, which significantly accelerates both the training and testing processes of our SOG-Net model.

In practice, these operators are efficiently implemented using existing packages. The gridding and 3D FFT operators can be performed by Type-1 non-uniform FFTs (NUFFTs)^{15,16}, while the 3D IFFT and gathering operators can be performed by Type-2 NUFFTs. The NUFFT is particularly effective for accelerating the exponential sums of the form $\sum_i w_i e^{i\mathbf{k}\cdot\mathbf{r}_i}$, where $\mathbf{k}$ and/or $\mathbf{r}_i$ exhibit non-uniform distributions. In this work, we utilize the FINUFFT package¹⁷ to accelerate the NUFFT computations. The error tolerance is set to 10^{-4} , and the so-called “exponential of semicircle” kernel is used as the shape function. The total computational complexity for each layer η is $O(N + N_{\text{FFT},\eta} \log N_{\text{FFT},\eta})$, where $N_{\text{FFT},\eta}$ denotes the size of the Fourier grid for layer η .

Algorithm 1 Accelerating the long-range convolution layer in the SOG-Net model

- Input:** Positions $\{\mathbf{r}_i\}$, the side length L , the shape function W , and the size of FFT grid $N_{\text{FFT},\eta}$, the latent variables $\{q_{i,\eta}\}$ and the SOG multipliers $\hat{g}_{\theta_\eta}$ at each layer $\eta = 1, \dots, P$.
- 1: (Gridding) For each layer η , spreading latent variables from atomic locations to a uniform grid to obtain $\Phi_{\text{grid},\eta}$ by using Eq. (III.3).
 - 2: (FFT) Obtain $\hat{\Phi}_{\text{grid},\eta}$ on the corresponding Fourier grid from $\Phi_{\text{grid},\eta}$ via a 3D FFT.
 - 3: (Diagonal Scaling) For each layer η , obtain $\hat{\Phi}_{\text{scal},\eta}$ by scaling $\hat{\Phi}_{\text{grid},\eta}$ with the SOG multiplier $\hat{g}_{\theta_\eta}$ and the $\hat{W}^{-2}$ term as in Eq. (III.4).
 - 4: (IFFT) Obtain $\Phi_{\text{scal},\eta}$ on real-space grids from $\hat{\Phi}_{\text{scal},\eta}$ via a 3D IFFT.
 - 5: (Gathering) For each layer η , obtain $\Phi_{i,\eta}$ by gathering contributions from grids as in Eq. (III.5).
- Output:** Long-range atomic energy $E^{LR,i} = \sum_{\eta=1}^P q_{i,\eta} \Phi_{i,\eta}$.
-

The LR component of the force, $\mathbf{F}^{LR,i}$, can be derived by applying automatic differentiation to E^{LR} . However, applying automatic differentiation for FFT and IFFT operators in the current implementation of the FINUFFT package is of rather expensive. We instead compute the gradient of E^{LR} with respect to the atomic position $\mathbf{r}_i$ directly using Eq. (I.5). The formula is expressed as:

$$\mathbf{F}^{LR,i} = -\nabla_{\mathbf{r}_i} E^{LR} = \frac{1}{2V} \sum_{\eta=1}^P q_{i,\eta} \sum_{\mathbf{k}} \mathbf{k} \hat{g}_{\theta_\eta}(\mathbf{k}) \text{Im} \left[e^{i\mathbf{k}\cdot\mathbf{r}_i} \sum_{j=1}^N q_{j,\eta} e^{-i\mathbf{k}\cdot\mathbf{r}_j} \right], \quad (\text{III.7})$$

where $\text{Im}[\cdot]$ denotes the imaginary part. In practice, the computation of Eq. (III.7) can be also accelerated by utilizing three additional Type-2 NUFFTs during the gathering step.

IV. TRAINING AND SIMULATION DETAILS

In this section, we provide details on the training and simulation setups used in our tests.

We start by describing the loss function and the error metrics used in our analysis. Given the importance of forces compared to energy in MD simulations, the loss function is defined as the mean squared error (MSE) of forces:

$$\mathcal{E}_{\text{loss}} := \frac{1}{N_{\text{batch}}} \sum_{n=1}^{N_{\text{batch}}} \sum_{i=1}^N \|\mathbf{F}_{\text{tot},n}^{\text{SOG-Net}}(\mathbf{r}_i) - \mathbf{F}_{\text{tot},n}^{\text{exact}}(\mathbf{r}_i)\|^2, \quad (\text{IV.1})$$

where n indexes the batches sampled from the training set in each epoch, N_{batch} is the total number of batches, and $\mathbf{F}_{\text{tot},n}^{\text{SOG-Net}}$ and $\mathbf{F}_{\text{tot},n}^{\text{exact}}$ represent the forces calculated by SOG-Net and the reference

solution derived from DFT, respectively. For tests involving binding energy curves of dimer pairs, we use the MSE of energies as the loss function; otherwise, we employ the above loss definition.

During testing, the relative root-mean-square error (R-RMSE) of forces is used as the error metric for the NaCl electrolyte model. It is defined as:

$$\mathcal{E}_{\text{R-RMSE}} := \sqrt{\frac{\sum_{m=1}^{N_{\text{test}}} \sum_{i=1}^N \|\mathbf{F}_{\text{tot},m}^{\text{exact}}(\mathbf{r}_i) - \mathbf{F}_{\text{tot},m}^{\text{SOG-Net}}(\mathbf{r}_i)\|^2}{\sum_{m=1}^{N_{\text{test}}} \sum_{i=1}^N \|\mathbf{F}_{\text{tot},m}^{\text{exact}}(\mathbf{r}_i)\|^2}}, \quad (\text{IV.2})$$

where m indexes the test configurations, and N_{test} is the size of testing set. For tests involving binding energy curves, the RMSE of energies is used as the error metric. In other tests, we calculate the RMSE of forces across all test configurations as:

$$\mathcal{E}_{\text{RMSE}} := \sqrt{\frac{1}{NN_{\text{test}}} \sum_{m=1}^{N_{\text{test}}} \sum_{i=1}^N \|\mathbf{F}_{\text{tot},m}^{\text{exact}}(\mathbf{r}_i) - \mathbf{F}_{\text{tot},m}^{\text{SOG-Net}}(\mathbf{r}_i)\|^2}. \quad (\text{IV.3})$$

We next outline the network architecture and training approaches used in our tests. For the NaCl electrolyte model tests, we use only the shell component of the SR network described in Section II, with cutoff radii set to $r_c = 0$ and $r_s = 2$ (adjusted as specified, if applicable). The sizes of the embedding network $f_{\theta_{emb,s}}$ and the fitting network $f_{\theta_{fit}}$ are configured as (2, 4, 8, 16, 32) and (32, 32, 32, 32, 32), respectively. The activation function after each layer is set as tanh to prevent gradient disappearance/explosion. In this system, all ions are modeled as soft spheres of diameter σ with σ the distance unit, that interact through a force field that includes both the Coulomb potential $1/r$ and the dispersive Lennard-Jones (LJ) potential. Since the charge strengths are explicitly defined for these tests, with Na and Cl atoms assigned +1 and -1 charges, respectively, the latent network is not required to generate latent variables from the SR descriptor. Parameters for the LR component of SOG-Net model are detailed in the main text. The models are trained using the Adam optimizer with an initial learning rate of 0.001, which decays by a factor of 0.97 every 10 epochs. The training process comprises four stages: starting with a batch size of 8 configurations and 200 epochs, with both the batch size and the number of epochs doubling at each subsequent stage.

For each classes of molecular pair in the dimer tests, we employ a core-shell structured hybrid SR descriptor in our SOG-Net model. The core and shell cutoff radii are set to $6\text{\AA}$ and $12\text{\AA}$, respectively, with scaling factors $j_c = 1/8$ and $j_s = 1/15$ for the core and shell components. The sizes of the embedding networks $f_{\theta_{emb,c}}$ and $f_{\theta_{emb,s}}$ are configured as (20, 40, 80), and the parameter $M_c^<$ for the core component is set as 80. The sizes of the fitting network $f_{\theta_{fit}}$ and the latent network $f_{\theta_{inn}}$ are set to (64, 64, 64) and (20, 40, 80, 1), respectively. Consequently, a single latent variable layer ($P = 1$) is sufficient for this set of tests. All networks uses the tanh activation function after each layer. Parameters for the LR component of the SOG-Net model are detailed in the main text. The models are trained using the Adam optimizer with an initial learning rate of 0.001, which decays by a factor of 0.98 every 10 epochs. The training process comprises of three stages: starting with a batch size of 2 configurations and 800 epochs, with both the batch size and the number of epochs doubling at each subsequent stage.

For the liquid water tests, we again use the core-shell structured SR descriptor in our SOG-Net model, with cutoff radii set to $4\text{\AA}$ and $7\text{\AA}$ for the core and shell components, respectively. The dimensions of embedding matrices and fitting matrix are set to (25, 50, 100) and (120, 120, 120),

respectively. The scaling factors j_c and j_s used in the core and shell components are set to $1/8$ and $1/14$. The parameter $M_c^<$ for the core component is set as 16. The sizes of the latent network are configured as (25, 50, 100, 16), so that the number of layer of latent variables is set as 16. All networks uses the tanh activation function after each layer. Parameters for the LR component of the SOG-Net model are detailed in the main text. The models are trained using the Adam optimizer with an initial learning rate of 0.001, which decays by a factor of 0.997 every 10 epochs. The training process comprises of two stages: starting with a batch size of 50 configurations and 10000 epochs, with both the batch size and the number of epochs doubling at the subsequent stage. The MD simulations are performed in the ASE package, and the LAMMPS interface implementation is our future work. To compute the charge structure factors, we are assuming that hydrogen atoms and oxygen atoms have charges of $+0.4238e$ and $-0.8476e$, respectively, similar to previous works¹⁸. This assumption only affects the absolute amplitude of the correlation function in Figure 4 of the main text, without altering the relative scale.

V. SUPPLEMENTARY RESULTS

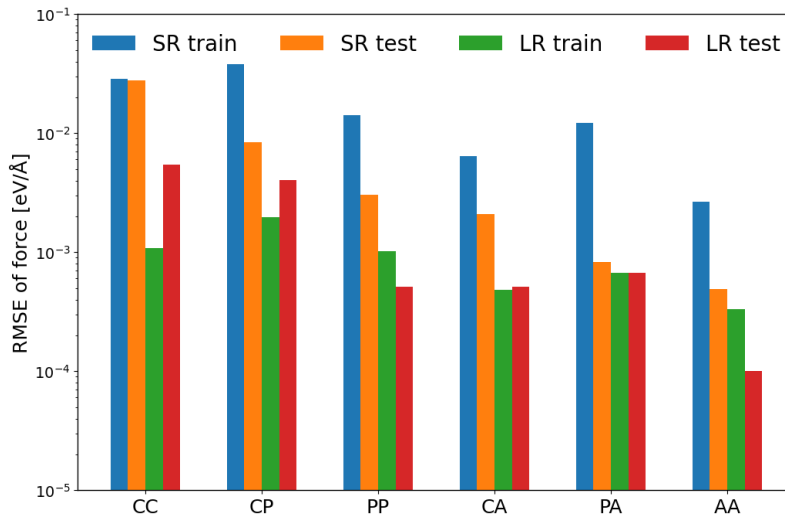


FIG. SI. Root mean square errors (RMSE) of forces for the test of dimer pairs. We report the training and testing errors using pure short-range (SR) and full SOG-Net (LR) models. For the SOG-Net model, the number of Fourier grids is set to 31 along each direction. Results are provided for charge-charge (CC), charge-polar (CP), polar-polar (PP), charge-apolar (CA), polar-apolar (PA), and apolar-apolar (AA) pairs.

¹K. K. Huguenin-Dumittan, P. Loche, N. Haoran, and M. Ceriotti, “Physics-inspired equivariant descriptors of nonbonded interactions,” *J. Phys. Chem. Lett.* **14**, 9612–9618 (2023).

²L. Zhang, H. Wang, M. C. Muniz, A. Z. Panagiotopoulos, R. Car, *et al.*, “A deep potential model with long-range electrostatic interactions,” *J. Chem. Phys.* **156**, 124107 (2022).

³T. W. Ko, J. A. Finkler, S. Goedecker, and J. Behler, “A fourth-generation high-dimensional neural network potential with accurate electrostatics including non-local charge transfer,” *Nat. Commun.* **12**, 398 (2021).

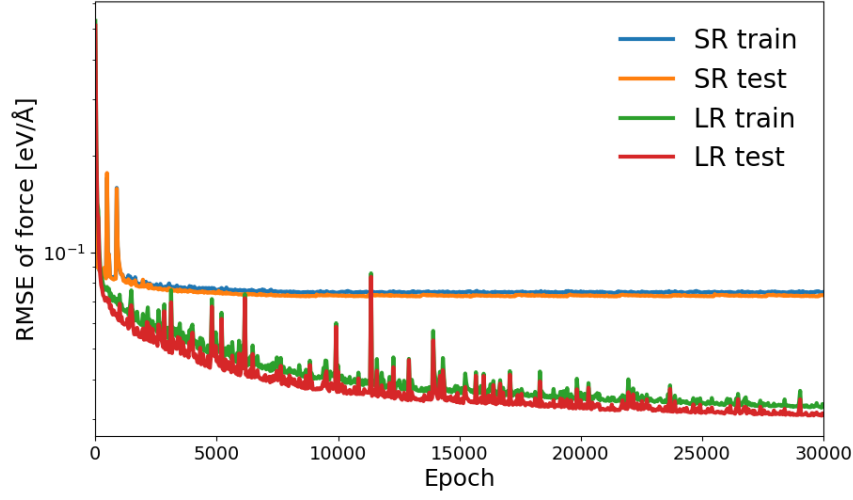


FIG. SII. Root-mean-square errors (RMSEs) of forces during the training process for liquid water tests. Results are presented for both the pure short-range (SR) component and the full SOG-Net model, evaluated on training and test datasets.

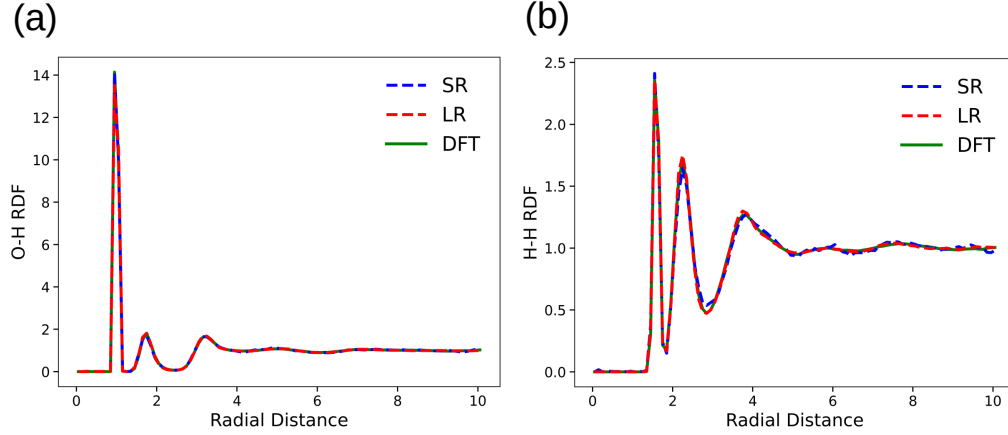


FIG. SIII. Predicted radial distribution functions (RDFs) of (a) oxygen-hydrogen (O-H) and (b) hydrogen-hydrogen (H-H) atom pairs of water at 300K and 1g/mL, using short-range (SR) or long-range (LR) models compared with the DFT results.

⁴Y. Shaidu, F. Pellegrini, E. Kckbenli, R. Lot, and S. de Gironcoli, “Incorporating long-range electrostatics in neural network potentials via variational charge equilibration from shortsighted ingredients,” *NPJ Comput. Mater.* **10**, 47 (2024).

⁵A. Kosmala, J. Gasteiger, N. Gao, and S. Gnnemann, “Ewald-based long-range message passing for molecular graphs,” in *International Conference on Machine Learning (ICML, 2023)* pp. 17544–17563.

⁶G. B. Folland and A. Sitaram, “The uncertainty principle: a mathematical survey,” *J. Fourier Anal. App.* **3**, 207–238 (1997).

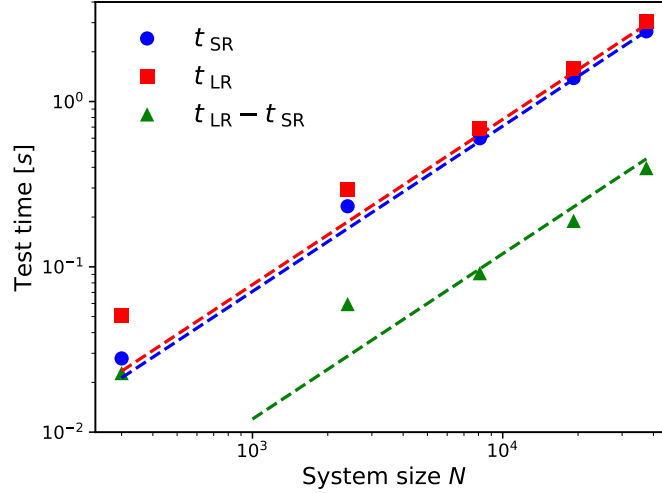


FIG. SIV. CPU time per step for the SOG-Net framework with increasing system size N while simulating liquid water systems. Data are shown for timing results obtained using the pure SR network (t_{SR}), the full SOG-Net (t_{LR}), and the additional cost beyond the pure SR model ($t_{\text{LR}} - t_{\text{SR}}$). The dashed lines show the linear fitting of data.

- ⁷C. Faller, M. Kaltak, and G. Kresse, “Density-based long-range electrostatic descriptors for machine learning force fields,” *J. Chem. Phys.* **161** (2024).
- ⁸A. P. Bartók, M. C. Payne, R. Kondor, and G. Csányi, “Gaussian approximation potentials: The accuracy of quantum mechanics, without the electrons,” *Phys. Rev. Lett.* **104**, 136403 (2010).
- ⁹P. Benner, S. Gugercin, and K. Willcox, “A survey of projection-based model reduction methods for parametric dynamical systems,” *SIAM Rev.* **57**, 483–531 (2015).
- ¹⁰L. Zhang, J. Han, H. Wang, R. Car, and W. E, “Deep potential molecular dynamics: a scalable model with the accuracy of quantum mechanics,” *Phys. Rev. Lett.* **120**, 143001 (2018).
- ¹¹A. V. Shapeev, “Moment tensor potentials: A class of systematically improvable interatomic potentials,” *Multiscale Model. Simul.* **14**, 1153–1173 (2016).
- ¹²I. Batatia, D. P. Kovacs, G. Simm, C. Ortner, and G. Csányi, “MACE: Higher order equivariant message passing neural networks for fast and accurate force fields,” *Adv. Neural Inf. Process. Syst. (NeurIPS)* **35**, 11423–11436 (2022).
- ¹³B. Cheng, “Cartesian atomic cluster expansion for machine learning interatomic potentials,” *NPJ Comput. Mater.* **10**, 157 (2024).
- ¹⁴A. Grisafi and M. Ceriotti, “Incorporating long-range physics in atomic-scale machine learning,” *J. Chem. Phys.* **151**, 204105 (2019).
- ¹⁵A. Dutt and V. Rokhlin, “Fast Fourier transforms for nonequispaced data,” *SIAM J. Sci. Comput.* **14**, 1368–1393 (1993).
- ¹⁶L. Greengard and J.-Y. Lee, “Accelerating the nonuniform fast Fourier transform,” *SIAM Rev.* **46**, 443–454 (2004).
- ¹⁷A. H. Barnett, J. Magland, and L. af Klinteberg, “A parallel nonuniform fast Fourier transform library based on an “exponential of semicircle” kernel,” *SIAM J. Sci. Comput.* **41**, C479–C504 (2019).
- ¹⁸B. Cheng, “Latent Ewald summation for machine learning of long-range interactions,” *arXiv:2408.15165* (2024).